

INDIAN STATISTICAL INSTITUTE

Practice Problems 30 July 2026
M.TECH(CS/CrS) I YEAR

INTRODUCTION TO PROGRAMMING

1. Read a predefined number of integers from the standard input (std in) and do the following:
 - (i) Print the sum of the numbers
 - (ii) Print the average of the numbers
 - (iii) Sort the numbers using insertion sort
 - (iv) Find the median of the numbers
2. Given an integer n , find the n -th Fibonacci number
3. Given two numbers find their GCD.
4. Recall that a permutation π of the integers $[n] = \{0, 1, 2, \dots, n-1\}$ may be represented by an array A of length n , by setting $A[i] = \pi(i)$. Write a C program which, given n and a list A of space-separated integers, determines
 - (a) whether A represents a permutation π of $[n]$, and
 - (b) prints π^{-1} in the same format if the answer to the above question is yes.